EtherChannel Questions

5 questions | Original English questions and source answers

Question 1

Refer to the exhibit.
Which two commands were used to create port channel 10? (Choose two)

```
Switch#show etherchannel summary
<output omitted>

Group  Port-channel  Protocol   Ports
------+-------------+----------+-------
10     Po10(SU)         LACP      Gi0/0(P)   Gi0/1(P)
20     Po20(SU)         LACP      Gi0/2(P)   Gi0/3(P)
```

Options

A. interface range g0/0-1
channel-group 10 mode on

B. interface range g0/0-1
channel-group 10 mode active

C. interface range g0/0-1
channel-group 10 mode auto

D. interface range g0/0-1
channel-group 10 mode passive

E. interface range g0/0-1
channel-group 10 mode desirable

Answer

B. interface range g0/0-1
channel-group 10 mode active

D. interface range g0/0-1
channel-group 10 mode passive

Source: CCNA Training » EtherChannel Quiz.pdf | Question 1

Question 2

Refer to the exhibit. Based on the LACP neighbor status, in which mode is the SW1 port channel configured?

```
SW1# show lacp neighbor
Flags:  S - Device is sending Slow LACPDUs
        F - Device is sending Fast LACPDUs
        A - Device is in Active mode
        P - Device is in Passive mode

Channel group 31 neighbors

Partner's information:

               LaCP port                        Admin  Oper  Port    Port
Port   Flags   Priority  Dev ID          Age    key    Key   Number  State
Et1/0  SP      32768     aabb.cc40.4000   6s    0x0    0x23  0x101   0x3C
Et1/1  SP      32768     aabb.cc50.4000   6s    0x0    0x23  0x101   0x3C
```

Options

A. active

B. mode on

C. auto

D. passive

Answer

A. active

Source explanation

From the neighbor status, we notice the “Flags” are SP. “P” here means the neighbor is in Passive mode. In order to create an Etherchannel interface, the (local) SW1 ports should be in Active mode. Moreover, the “Port State” in the exhibit is “0x3c” (which equals to “00111100″ in binary format). Bit 3 is “1” which means the ports are synchronizing -> the ports are working so the local ports should be in Active mode.
Reference:
https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3650/software/release/3se/consolidated_guide/comm

Source: CCNA Training » EtherChannel Quiz.pdf | Question 2

Question 3

Which two command sequences must you configure on a switch to establish a Layer 3 EtherChannel with an open-standard protocol? (Choose two)

Options

```
A. interface GigabitEthernet0/0/1
channel-group 10 mode auto
```

```
B. interface port-channel 10
switchport
switchport mode trunk
```

```
C. interface GigabitEthernet0/0/1
channel-group 10 mode active
```

```
D. interface port-channel 10
no switchport
ip address 172.16.0.1 255.255.255.0
```

```
E. interface GigabitEthernet0/0/1
channel-group 10 mode on
```

Answer

C. interface GigabitEthernet0/0/1
channel-group 10 mode active

D. interface port-channel 10
no switchport
ip address 172.16.0.1 255.255.255.0

Source explanation

In order to configure a Layer 3 EtherChannel, we must use the “no switchport” command and set the IP address
on the (logical) port-channel interface. For open-standard EtherChannel protocol we must use LACP with “passive” & “active” mode.

Source: CCNA Training » EtherChannel Quiz.pdf | Question 3

Question 4

Which mode must be used to configure EtherChannel between two switches without using a negotiation protocol?

Options

A. active

B. auto

C. desirable

D. on

Answer

D. on

Source explanation

The Static Persistence (or “on” mode) bundles the links unconditionally and no negotiation protocol is used. In this mode, neither PAgP nor LACP packets are sent or received.

Source: CCNA Training » EtherChannel Quiz.pdf | Question 4

Question 5

Which two statements about EtherChannel technology are true? (Choose two)

Options

A. EtherChannel does not allow load sharing of traffic among the physical links within the EtherChannel

B. EtherChannel provides increased bandwidth by bundling existing FastEthernet or Gigabit Ethernet interfaces into a single EtherChannel

C. STP does not block EtherChannel links

D. You can configure multiple EtherChannel links between two switches, using up to a limit of sixteen physical ports

E. EtherChannel allows redundancy in case one or more links in the EtherChannel fail

Answer

B. EtherChannel provides increased bandwidth by bundling existing FastEthernet or Gigabit Ethernet interfaces into a single EtherChannel

E. EtherChannel allows redundancy in case one or more links in the EtherChannel fail

Source explanation

In fact answer 'STP does not block EtherChannel links' is also correct as STP considers Etherchannel links as one physical link so this question should have three correct answers. But answer 'STP does not block EtherChannel links' is not as obvious as the other two answers so we should choose them.
Answer "You can configure multiple EtherChannel links between two switches, using up to a limit of sixteen physical ports" is not correct as "Commonly up to 16 links can be assigned to an EtherChannel, but only 8 can be active at a time".
Reference: https://www.ciscopress.com/articles/article.asp?p=2348266&seqNum=3
So it should be 8 physical ports, not 16.

Source: CCNA Training » EtherChannel Quiz.pdf | Question 5